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Optical apparatus, evaluation apparatus, evaluation method, and manufacturing method of optical system

Abstract

An optical apparatus includes a first light source, a second light source, a chart, an optical system, and a light receiving system. The chart is configured to guide light emitted from the first light source to a target optical system. The optical system is configured to form a point image by using light emitted from the second light source. The light receiving system is configured to receive first light emitted from the chart via the target optical system and second light emitted from the point image via the target optical system. The first light and the second light enter different positions of the target optical system.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The aspect of the embodiments relates to an optical apparatus, an evaluation apparatus, an evaluation method, and a manufacturing method of an optical system.

Description of the Related Art

(2) Japanese Patent Laid-Open No. (“JP”) 2007-212620 discloses a method that measures modulation transfer functions (MTF) at a plurality of image heights by using light emitted from a chart irradiated by a light source. By using this method, it is possible to acquire a one-side blur amount from MTFs and to adjust an image pickup optical system so that the one-side blur amount is reduced.

(3) JP 2019-066428 discloses a method in which light emitted from a high-intensity point light source is transmitted through an image pickup optical system and a wavefront sensor acquires a transmitted wavefront. By using this method, it is possible to acquire basic aberrations of an image pickup optical system such as spherical aberration, coma, astigmatism, field curvature, and one-side blur from on-axis and off-axis wavefront aberrations.

(4) The method disclosed in JP 2007-212620 can measure the MTFs at the plurality of image heights of various optical systems, but can acquire only part (one-side blur and field curvature) of aberration (optical performance). The method disclosed in JP 2019-066428 can measure on-axis and off-axis wavefront aberrations, but when the method is to perform the measurement on image pickup optical systems having different specifications such as angles of views and sensor sizes, it is necessary to change a position of the sensor and the light source.

SUMMARY OF THE INVENTION

(5) The present disclosure provides an optical apparatus, an evaluation apparatus, an evaluation method, and a manufacturing method of an optical system each of which can easily evaluate optical performance of various optical systems.

(6) An optical apparatus according to one aspect of the embodiments includes a first light source, a second light source, a chart, an optical system, and a light receiving system. The chart is configured to guide light emitted from the first light source to a target optical system. The optical system is configured to form a point image by using light emitted from the second light source. The light receiving system is configured to receive first light emitted from the chart via the target optical system and second light emitted from the point image via the target optical system. The first light and the second light enter different positions of the target optical system.

(7) An evaluation apparatus according to one aspect of the embodiments includes the optical apparatus, and at least one processor or circuit. The at least one processor or circuit is configured to execute a plurality of tasks including a controlling task. The controlling task is configured to evaluate optical performance of the target optical system by using output from the light receiving system.

(8) An evaluation method according to one aspect of the embodiments is an evaluation method for optical performance of a target optical system. The evaluation method includes irradiating a chart including a plurality of patterns by using a first light source, receiving light having been emitted from the chart and transmitted through the target optical system, calculating a frequency response characteristic of the target optical system at positions of the plurality of patterns by using the received light, shaping light emitted from a second light source different from the first light source to be a point image at a position different from the positions of the plurality of patterns, causing a wavefront sensor to receive light having been emitted from the point image and transmitted through the target optical system, and calculating wavefront aberration of the target optical system by using the light received by the wavefront sensor.

(9) An evaluation method according to one aspect of the embodiments is an evaluation method for optical performance of a target optical system. The evaluation method includes irradiating a chart

including a plurality of patterns by using a first light source, receiving light having been emitted from the chart and transmitted through the target optical system, calculating a frequency response characteristic of the target optical system at positions of the plurality of patterns by using the received light, shaping light emitted from a second light source different from the first light source to be a point image at a position different from the positions of the plurality of patterns, causing a light intensity sensor to receive, at a plurality of focus positions, light having been emitted from the point image and transmitted through the target optical system, and calculating wavefront aberration of the target optical system by using a plurality of lights received at the plurality of focus positions.
(10) A manufacturing method according to one aspect of the embodiments is a manufacturing method of an optical system including a plurality of lenses. The manufacturing method includes manufacturing the plurality of lenses, assembling the plurality of lenses, evaluating optical performance by using one of the evaluation methods, and modifying at least part of the assembled plurality of lenses based on the evaluated optical performance.
(11) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic diagram illustrating an evaluation apparatus according to a first embodiment.
(2) FIG. **2** is an explanatory diagram illustrating an SHS according to the first embodiment.
(3) FIG. **3** is a configuration diagram illustrating a chart according to the first embodiment.
(4) FIGS. **4**A to **4**C are diagrams illustrating patterns of the chart according to the first embodiment.
(5) FIG. **5** is an explanatory diagram illustrating an evaluation apparatus according to the first embodiment.
(6) FIG. **6** is a schematic diagram illustrating an evaluation apparatus according to a second embodiment.
(7) FIG. **7** is a schematic diagram illustrating an evaluation apparatus as a modification example of the second embodiment.
(8) FIG. **8** is a flow chart illustrating a manufacturing method of an optical system according to a third embodiment.
DESCRIPTION OF THE EMBODIMENTS
(9) Referring now to the accompanying drawings, a description is given of embodiments according to the present disclosure.
First Embodiment
(10) First, with reference to FIG. **1**, a description is given of an evaluation apparatus (measuring apparatus) according to a first embodiment of the present disclosure. FIG. **1** is a schematic diagram illustrating an evaluation apparatus **100** according to this embodiment. The evaluation apparatus **100** evaluates optical performance of a target optical system, and in the evaluation apparatus **100**, light emitted from a light emitting system is transmitted through the target optical system and then the light receiving system receives the light. The evaluation apparatus **100** can evaluate (measure) on-axis transmitted wavefronts and off-axis MTFs of various target optical systems.
(11) The evaluation apparatus **100** includes a PC (controlling unit or controlling task) **160** and an optical apparatus having a light emitting system that emits light to a target lens (target optical system) **130** and a light receiving system that receives light from the target lens **130**. The light emitting system includes a laser light source (second light source) **110**, a fiber **111**, collimator lenses **112** and **113**, a ring light source (first light source) **120**, a chart **121**, and a stage **122**. The

ring light source **120** has a ring shape and has an opening through which light from the laser light source **110** is transmitted. The light receiving system includes an intensity sensor (light intensity sensor) **140**, a collimator lens **150**, and a Shack-Hartmann sensor (SHS) **151** as a wavefront sensor. The PC **160** controls each part of the optical apparatus. The target lens **130** is an image pickup optical system including a plurality of optical elements, and is disposed at a position such that a relationship between the chart **121** and the intensity sensor **140** is a relationship between an object and an image.
(12) First, a description is given of a system for measuring an on-axis transmitted wavefront of the target lens **130**. Light **10** emitted from the laser light source **110** via a fiber **111** is made to be parallel light by the collimator lens **112** and is focused on a focal position of the target lens **130** via the collimator lens **113**. The focused light becomes diverging light **11** after becoming a simulated point light source at the focal position of the target lens **130**, and enters the target lens **130**. The diverging light **11** becomes a converging light **12** via the target lens **130**, becomes a diverging light again after being focused, and enters the collimator lens **150**. The light that is made to be approximately parallel light by the collimator lens **150** is imaged by the SHS **151**. A signal acquired by the SHS **151** is calculated by the PC **160**.
(13) Next, with reference to FIG. **2**, a description is given of a case where a wavefront of entering light is measured by the SHS **151**. FIG. **2** is an explanatory diagram of the SHS **151**, and illustrates a sectional view of the SHS **151** when the SHS **151** measures a wavefront of the entering light. The entering light is divided by a microlens array **152** in FIG. **2** and focused on a CMOS sensor **153** disposed on a rear side. The light focused on the CMOS sensor **153** is analyzed by the PC **160** illustrated in FIG. **1**. Tilt of the wavefront of the entering light is calculated from a position of a center of gravity of the light-focused point, and a shape of the wavefront of the entering light is acquired by integrating or fitting data group of the tilt of the wavefront.
(14) In order that the transmitted wavefront of the target lens **130** is accurately acquired, the wave entering the target lens **130** is made to be an ideal spherical wave emitted from a point light source having a size smaller than the following equation (1). In the equation (1), λ represents a wavelength of the laser light source **110**, and NA represents a numerical aperture of the target lens **130**.
(15) $\sim\frac{}{NA}$ (1)
(16) Forming the point light source at the focal position of the target lens **130** requires an NA of the collimator lens **113** to he greater than the NA of the target lens **130**. The laser light source **110** is a high-intensity light source capable of emitting light having a coherence that can converge into a size smaller than the size expressed by the equation (1) and having a light amount that can be measured by the SHS **151**. Such a light source includes, for example, a gas laser and a semiconductor laser. Light acquired by focusing light having a high coherence with a sufficiently large NA can be regarded as a simulated point light source with respect to the target lens **130**.
(17) Next, a description is given of a system for measuring an off-axis MTF of the target lens **130**. The light emitted from the ring light source **120** illustrated in FIG. **1** passes through a transmitting portion of the chart **121** and enters the target lens **130** as diverging lights **21** and **31**. The diverging lights **21** and **31** become converging lights **22** and **32** after passing through the target lens **130**, and the intensity sensor **140** receives them. The intensity sensor **140** can be driven by an unillustrated stage.
(18) FIG. **3** is a configuration diagram of the chart **121**. At the center of the chart **121**, an opening **1211** is formed for transmitting light emitted from the laser light source **110**, and the light emitted from the laser light source **110** passes through the opening **1211**. An absorbing portion **1212** and a transmitting portion **1213** are provided in a peripheral part of the chart **121**, and a diffuser plate **1214** is disposed on an upstream side of the transmitting portion **1213**. The diffuser plate **1214** is provided to make uniform light distribution characteristics of the light emitted from the ring light source **120**. Thus, in this embodiment, the chart **121** includes a plurality of transmitting portions

1213 and a plurality of absorbing portions **1212**, and the point light source formed by the laser light source **110** is located at one (first transmitting portion) of the plurality of transmitting portions **1213** (opening **1211**). In this embodiment, the diffuser plate **1214** is disposed between a transmitting portion (second transmitting portion) among the plurality of transmitting portions **1213** and the ring light source **120**, and the point light source is not located on the transmitting portion. In this embodiment, the transmitting portion **1213** is air but is not limited to this, and a transmitting member that transmits light may be disposed. In this embodiment, the transmitting portions **1213** of the chart **121** are collectively referred to as a chart pattern or pattern.
(19) In a case where the light emitted from the ring light source **120** overlaps the diverging light **11** from the laser light source **110**, a measurement error may occur. In order that the overlap is avoided, a light blocking portion **1215** may be provided as illustrated in FIG. **3**. However, this embodiment is not limited to this, and a light blocking portion may be located at a position other than a position on the chart **121**, as long as the configuration can avoid the overlap between the light emitted from the ring light source **120** and the diverging light **11** from the laser light source **110**. Alternatively, without the light blocking portion disposed, the PC **160** may turn off the ring light source **120** while the SHS **151** is receiving the light. As a result, both lights can be separated from each other even in a case where a light blocking portion is not provided.
(20) FIGS. **4**A to **4**C are diagrams illustrating patterns of the chart **121**. FIG. **4**A illustrates an edge pattern, FIG. **4**B illustrates a slit pattern, and FIG. **4**C illustrates a grating pattern. The MTF value can be calculated from a degree of blurring of each chart image. A defocus characteristic of the MTF value can also be acquired by the stage **122** making the chart **121** defocused.
(21) As described above, the light emitting system includes a first light emitting system that forms the light emitted from the chart **121** illuminated (irradiated) by the ring light source **120**, and a second light emitting system that shapes the light emitted from the laser light source **110** to form the point light source. The intensity sensor **140** receives the light having been emitted from the first light emitting system and transmitted through the target lens **130**, and the SHS **151** receives the light having been emitted from the second light emitting system and transmitted through the target lens **130**.
(22) These two types of light emitting systems enable the acquisition of the on-axis transmitted wavefront and the off-axis MTF of the target lens **130**. With respect to the on-axis transmitted wavefront, for example, a wavefront shape can be decomposed into aberration components by fitting the on-axis transmitted wavefront with a Zernike polynomial. Fitted Zernike coefficients include information similar to basic aberration such as astigmatism, coma, and spherical aberration, as described in the following equations (2).
Term Z4=Defocus Component
Term Z5=Astigmatism Component at 0°-90°
Term Z6=Astigmatism Component at 45°
Term Z7=Horizontal Coma Component
Term Z8=Vertical Coma Component
Term Z9=Spherical Aberration Component (2)
(23) One-side blur amount of the target lens **130** can be calculated from a variation in peak positions of the off-axis MTFs, and a field curvature amount of the target lens **130** can be calculated from the peak position of the off-axis MTF and the defocus term of the Zernike polynomial. In this way, it is possible to acquire the basic aberration of the target lens **130**.
(24) Next, with reference to FIG. **5**, a description is given of a case where the evaluation apparatus **100** performs a measurement on a target lens (target optical system) **131** having different optical specifications. FIG. **5** is an explanatory diagram of the evaluation apparatus **100**. For example, in a case where target lenses **131** have different focal lengths, the respective distances from target lenses **131** to the intensity sensor **140** or the SHS **151** are adjusted. In a case where target lenses **131** having different angles of views are to be measured, the intensity sensor **140** is moved so that the

intensity sensor **140** is located at respective positions of the angles of views calculated from respective image heights and focal lengths. In a case where MTFs of different image heights are to be measured, the intensity sensor **140** is positioned so that the intensity sensor **140** measures light emitted from patterns at different image heights, such as light **23** and **24** and light **33** and **34** in FIG. **5**. In this case, the pattern of the chart **121** may be located at a plurality of image heights as illustrated in FIG. **4**A so that the same chart **121** can be used when the target lenses **131** is replaced.
(25) As described above, the optical apparatus according to this embodiment includes the first light source (ring light source **120**), the chart **121** that guides the light emitted from the first light source to the target optical system (target lens **130**), and the second light source (laser light source **110**) that is different from the first light source. The optical apparatus also includes the optical system (fiber **111**, collimator lenses **112** and **113**) that forms the point image by using the light emitted from the second light source. The optical apparatus further includes the light receiving system (intensity sensor **140**, SHS **151**) that receives first light emitted from the chart via the target optical system and second light emitted from the point image via the target optical system. The first light and the second light enter different positions of the target optical system.
(26) The evaluation apparatus **100** according to this embodiment includes the optical apparatus and the controlling unit (controlling task, PC **160**) that evaluates the optical performance of the target optical system by using the output from the light receiving system. Here, the different positions of the target optical system are, for example, on-axis (center) and off-axis (periphery), but the laser beam does not have to be on-axis. The optical performance may be a frequency response characteristic (MTF) at a position of the chart of the first light emitting system and wavefront aberration at a position of the point light source of the second light emitting system.
(27) The evaluation apparatus **100** includes the light intensity sensor that receives the first light and the wavefront sensor that receives the second light, and executes causing the wavefront sensor to receive the light that has been emitted from the point image and transmitted through the target optical system, and calculating the wavefront aberration of the target optical system by using the light received by the wavefront sensor. However, this embodiment is not limited to this, and the evaluation apparatus may include a light intensity sensor that receives the first light and the second light. In this case, the evaluation apparatus executes causing the light intensity sensor to receive, at a plurality of focus positions, the light that has been emitted from the point image and transmitted through the target optical system, and calculating the wavefront aberration of the target optical system by using the light received at the plurality of focus positions.
(28) According to this embodiment, it is possible to provide an optical apparatus, an evaluation apparatus, and an evaluation method each of which can easily evaluate optical performance of various optical systems.
Second Embodiment
(29) Next, with reference to FIG. **6**, a description is given of an evaluation apparatus (measuring apparatus) according to a second embodiment of the present disclosure. FIG. **6** is a schematic diagram illustrating an evaluation apparatus **200** according to this embodiment.
(30) The evaluation apparatus **100** includes an unillustrated PC (controlling unit, controlling task) and an optical apparatus including a light emitting system that emits light to a target lens (target optical system) **132** and a light receiving system that receives light from the target lens **132**. The light emitting system includes an unillustrated laser light source (second light source), a fiber **111**, collimator lenses **112** and **113**, a chart **123**, a beam splitter **201**, a reflection mirror **203**, and an LED light source (first light source) **204**. The LED light source **204** includes a plurality of LEDs arranged so that an opening is formed for transmitting light emitted from the laser light source. The light receiving system includes a wavefront sensor **202**, a collimator lens **205**, and an intensity sensor (light intensity sensor) **206**. The target lens **132** is an image pickup optical system including a plurality of optical elements, and is disposed at a position such that a relationship between the chart **123** and an object at an infinite distance are a conjugated relationship.

(31) Light emitted from the unillustrated laser light source through the fiber **111** forms a point light source (simulated point light source) at an opening of the chart **123**, as in the first embodiment. The light emitted from the simulated point light source is transmitted through the target lens **132**, is reflected by a reflection mirror **203**, and then is returned to the collimator lens **113** through the same path. The light returned to the collimator lens **113** is then reflected by the beam splitter **201** and acquired by the wavefront sensor **202**. The wavefront sensor **202** is, for example, an SHS, but is not limited to this, and may be a shearing interferometer or a talbot interferometer.
(32) The light emitted from the LED light source **204** illuminates (irradiate) the chart **123** via a light guide including a fiber or the like. The illumination light passes through the transmitting portion of the chart **123**, passes through the target lens **130**, becomes approximately parallel light, passes through the collimator lens **205**, and is then received by the intensity sensor **206**. The light guide may be prepared for each transmitting portion of the chart **123**, or one light guide may illuminate (irradiate) a plurality of transmitting portions.
(33) As in the chart **121**, the chart **123** includes a diffuser plate located on a pattern for MTF measurement. The opening of the chart **123** does not need to be located on an optical axis, as long as the opening is located on an image plane of the target lens **132**. In a case where the opening is not located on the optical axis, an exit pupil of the target lens **132** may not have a circular shape. In this case, a coefficient may be calculated for an elliptical shape using an elliptical Zernike. Alternatively, the coefficient may be acquired by using an orthogonalization function that matches a shape of the exit pupil based on a Gram-Schmitt concept.
(34) In a case where the chart **123** has a grating pattern, the intensity sensor **206** always measures a contrast of the pattern at a specific frequency and at a specific defocus. By tilting the pattern or the intensity sensor, the defocus characteristic can be acquired in an area of the intensity sensor **206**.
(35) In the evaluation apparatus **200**, in a case where the light from the LED light source **204** enters the wavefront sensor **202**, an error occurs. In order that the error is reduced, a diffuser plate may be placed at a position away from the opening so that diffused light from the diffuser plate does not enter the opening. Alternatively, a light blocking portion that surrounds the optical path of the laser beam may be provided. That is, at least one of the first light emitting system and the second light emitting system may include a light blocking portion that separates the light from the first light emitting system and the light from the second light emitting system. Alternatively, the LED light source **204** may be turned off while the wavefront sensor **202** is receiving light. In a case where a light blocking portion surrounding the optical path of the laser beam is provided between the collimator lens **113** and chart **123**, an outer side of the light blocking portion may be made to be a mirror surface, and the light emitted from the LED light source **204** may he reflected, which improves a light utilization efficiency.
(36) Next, with reference to FIG. **7**, a description is given of a modification example of the evaluation apparatus according to this embodiment. FIG. **7** is a schematic diagram illustrating the evaluation apparatus **210** as a modification example of this embodiment. The evaluation apparatus **210** includes a fiber **111**, a chart **124**, an intensity sensor (light intensity sensor) **211**, and a stage **212**. A target lens **133** is an image pickup optical system including a plurality of optical elements, and is disposed at a position such that a relationship between the chart **124** and the intensity sensor **211** is a relationship between an object and an image.
(37) The chart **124** is a black-and-white chart including an absorbing portion (black) and a scattering portion (white), and is illuminated (irradiated) by an unillustrated light source (first light source). The first light source is an illumination light source such as a fluorescent lamp. The scattered light from the pattern on the chart **124** is projected onto part of a screen of the intensity sensor **211**. Light emitted from an unillustrated second light source is guided by the fiber **111** and projected onto another part of the screen of the intensity sensor **211**. The intensity sensor **211** is driven in the optical axis direction by the stage **212** so that the intensity sensor **211** acquires a plurality of defocus images. Defocus characteristics of MTFs can be calculated from the defocus

images of the chart pattern. A transmitted wavefront of the target lens **133** can be calculated from the defocus images of the light emitted from the fiber **111** by an optimization method.
(38) The fiber **111** may be disposed between the chart **124** and the target lens (target optical system) **133**. In this case, an absorbing portion (black) is disposed on an upstream side of the fiber **111** so that the light from the scattering portion (white) of the chart **124** does not pass through the vicinity of a light emitting position of the fiber **111** and the light emitted from the fiber **111** and the light from the scattering portion (white) do not overlap each other on the intensity sensor **211**. A size D of the absorbing portion where the point image is located is determined based on an aperture size of the target lens **133** and a defocus amount of the intensity sensor **211** of the light receiving system. More specifically, the following equation (3) expresses D using a defocus amount Z of the intensity sensor **211**, a numerical aperture Fno of the target lens **133**, and an object distance m.
$D=mZ/Fno$ (3)
(39) Generally, the object distance m is 10 or more. That is, the measurement is performed at an object distance m that is 10 times or more the focal length f of the target lens (target optical system) **133**. A maximum value of the defocus amount Z is 0.1 Fno to 2 Fno [mm], and hence the size D of the absorbing portion may be at least 1 mm, or may be 20 mm or more.
Third Embodiment
(40) Next, with reference to FIG. **8**, a description is given of a manufacturing method of an optical system according to a third embodiment of the present disclosure. FIG. **8** is a flowchart illustrating the manufacturing method of the optical system according to this embodiment.
(41) First, in step S**301**, a single lens is manufactured by polishing a lens (lens manufacturing process). Subsequently, in step S**302**, a lens unit is assembled by combining single lenses (lens assembling process). Alternatively, an image pickup optical system (target optical system) is assembled by combining a single lens(es) and/or a lens unit(s). Subsequently, in step S**303**, basic aberration of the image pickup optical system (target optical system) is measured and evaluated (evaluating process). Subsequently, in step S**304**, it is determined whether the measured value is better (OK) than a reference or not (NG). In a case of OK, this flow ends. On the other hand, in a case of NG, the image pickup optical system (target optical system) is modified (corrected) in step S**305**. Then, the process returns to step S**303** again.
(42) In step S**305**, the target optical system is modified by adjusting an adjustment point(s) provided in the target optical system. In a case where the modification (correction) cannot be completed only by adjusting the adjustment point, the target optical system is disassembled, a point other than the adjustment point is adjusted, and the target optical system is reassembled. Alternatively, a lens or some lenses of the target optical system is modified by polishing the lens. Alternatively, a defective unit or lens is replaced. By going through the above steps, a high-quality optical system can be manufactured.
(43) According to each embodiment, it is possible to provide an optical apparatus, an evaluation apparatus, an evaluation method, and a manufacturing method of an optical system, each of which can easily evaluate the optical performance of various optical systems.
(44) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
(45) This application claims the benefit of Japanese Patent Application No. 2021-153672, filed on Sep. 21, 2021, which is hereby incorporated by reference herein in its entirety.

Claims

1. An optical apparatus comprising: a first light source and a second light source; a chart configured to guide light emitted from the first light source to a target optical system, the chart comprising: a

light blocking portion surrounding an aperture, wherein the light blocking portion is disposed between the aperture and a diffuser plate, wherein the light blocking portion is configured to separate the light emitted from the first light source and light emitted from the second light source, wherein an opening is formed on the diffuser plate, and wherein the light emitted from the second light source is transmitted through the opening; an optical system configured to form a point image by using the light emitted from the second light source; and a light receiving system configured to receive first light emitted from the chart via the target optical system and second light emitted from the point image via the target optical system, wherein the first light and the second light enter different positions of the target optical system, and wherein the size of the point image is smaller than λ/NA, where NA is a numerical aperture of the light receiving system and λ is a wavelength of the second light.

2. The optical apparatus according to claim 1, wherein the light receiving system includes: a light intensity sensor configured to receive the first light; and a wavefront sensor configured to receive the second light.

3. The optical apparatus according to claim 1, wherein the light receiving system includes a light intensity sensor configured to receive the first light and the second light.

4. The optical apparatus according to claim 1, wherein the chart includes a plurality of scattering portions and a plurality of absorbing portions, and wherein the point image is located on an absorbing portion among the plurality of absorbing portions.

5. The optical apparatus according to claim 4, wherein a size of the absorbing portion on which the point image is located is determined based on an aperture size of the target optical system and a defocus amount of a light intensity sensor of the light receiving system.

6. The optical apparatus according to claim 1, wherein the chart includes a first transmitting portion and an absorbing portion, and wherein the point image is located on the first transmitting portion.

7. The optical apparatus according to claim 6, wherein the chart includes a second transmitting portion on which the point image is not located, and wherein the diffuser plate is disposed between the second transmitting portion and the first light source.

8. The optical apparatus according to claim 1, wherein the second light source is a laser light source.

9. The optical apparatus according to claim 1, wherein an opening is formed on the first light source, and wherein the light emitted from the second light source is transmitted through the opening formed on the first light source.

10. The optical apparatus according to claim 9, wherein the first light source is a ring light source.

11. The optical apparatus according to claim 9, wherein the first light source is an LED light source including a plurality of LEDs.

12. An evaluation apparatus comprising: an optical apparatus according to claim 1; a memory storing instructions; and at least one processor that executes the instructions to evaluate optical performance of the target optical system by using output from the light receiving system.

13. The evaluation apparatus according to claim 12, wherein the optical performance includes a frequency response characteristic at a position of the chart and wavefront aberration at a position of the point image.

14. The evaluation apparatus according to claim 12, wherein the at least one processor executes the instructions to turn-off the first light source while the light receiving system is receiving the second light.

15. An evaluation method for optical performance of a target optical system, the evaluation method comprising: irradiating a chart including a plurality of patterns by using a first light source; receiving light having been emitted from the chart and transmitted through the target optical system; calculating a frequency response characteristic of the target optical system at positions of the plurality of patterns by using the received light; shaping light emitted from a second light source different from the first light source to be a point image at a position different from the

positions of the plurality of patterns; causing a wavefront sensor to receive light having been emitted from the point image and transmitted through the target optical system; and calculating wavefront aberration of the target optical system by using the light received by the wavefront sensor, wherein the size of the point image is smaller than λ/NA, where NA is a numerical aperture of a light receiving system and λ is a wavelength of the second light, wherein the chart comprises a light blocking portion surrounding an aperture, wherein the light blocking portion is disposed between the aperture and a diffuser plate, wherein the light blocking portion is configured to separate light emitted from the first light source and the light emitted from the second light source, wherein an opening is formed on the diffuser plate, and wherein the light emitted from the second light source is transmitted through the opening.

16. An evaluation method for optical performance of a target optical system, the evaluation method comprising: irradiating a chart including a plurality of patterns by using a first light source; receiving light having been emitted from the chart and transmitted through the target optical system; calculating a frequency response characteristic of the target optical system at positions of the plurality of patterns by using the received light; shaping light emitted from a second light source different from the first light source to be a point image at a position different from the positions of the plurality of patterns; causing a light intensity sensor to receive, at a plurality of focus positions, light having been emitted from the point image and transmitted through the target optical system; and calculating wavefront aberration of the target optical system by using a plurality of lights received at the plurality of focus positions, wherein the size of the point image is smaller than λ/NA, where NA is a numerical aperture of a light receiving system and λ is a wavelength of the second light, wherein the chart comprises a light blocking portion surrounding an aperture, wherein the light blocking portion is disposed between the aperture and a diffuser plate, wherein the light blocking portion is configured to separate light emitted from the first light source and the light emitted from the second light source, wherein an opening is formed on the diffuser plate, and wherein the light emitted from the second light source is transmitted through the opening.

17. A manufacturing method of an optical system including a plurality of lenses, the manufacturing method comprising: manufacturing the plurality of lenses; assembling the plurality of lenses into the optical system; evaluating optical performance of the optical system by using the evaluation method according to claim 15; and modifying at least part of the optical system-based on the evaluated optical performance.

18. A manufacturing method of an optical system including a plurality of lenses, the manufacturing method comprising: manufacturing the plurality of lenses; assembling the plurality of lenses into the optical system; evaluating optical performance of the optical system by using the evaluation method according to claim 16; and modifying at least part of the optical system based on the evaluated optical performance.